Littlewood–Richardson Coefficients

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Abstract

We give a self- contained, rigorous exposition of a Python program that computes the $(k{+}1)^{\mathrm{st}}$ socle layer of a tensor product of induced modules. The algorithm relies on combinatorial representation theory: partitions, skew shapes, Littlewood–Richardson (LR) coefficients, and Yamanouchi (lattice) words. All mathematical notions are introduced together with the corresponding code fragments; the overall flow of the program is described in detail.

Contents

Table 1: Table for $\lambda = (3,), \mu = (4,), k = 6$

$k = 0: \quad V^{(3),(4)}$
$k = 1: \quad V^{(2),(3)} \oplus V^{(2),(4)} \oplus V^{(3),(3)}$
$k = 2: \quad 2V^{(1),(2)} \oplus 2V^{(1),(3)} \oplus V^{(1),(4)} \oplus 2V^{(2),(2)} \oplus 2V^{(2),(3)} \oplus V^{(3),(2)}$
$k = 3:$ $V^{\varnothing,(1)} \oplus V^{\varnothing,(2)} \oplus V^{\varnothing,(3)} \oplus 6V^{(1),(1)} \oplus 5V^{(1),(2)} \oplus 2V^{(1),(3)} \oplus 4V^{(2),(1)} \oplus 2V^{(2),(2)} \oplus V^{(3),(1)}$
$k = 4:$ $V^{\varnothing,\varnothing} \oplus 5V^{\varnothing,(1)} \oplus 4V^{\varnothing,(2)} \oplus V^{\varnothing,(3)} \oplus V^{(1),\varnothing} \oplus 10V^{(1),(1)} \oplus 4V^{(1),(2)} \oplus V^{(2),\varnothing} \oplus 2V^{(2),(1)}$
$k = 5: \quad 7V^{\varnothing,\varnothing} \oplus 9V^{\varnothing,(1)} \oplus 2V^{\varnothing,(2)} \oplus 6V^{(1),\varnothing} \oplus 4V^{(1),(1)} \oplus V^{(2),\varnothing}$

Table 2: Table for $\lambda = (3,), \mu = (3,1), k = 6$

$k = 0: \quad V^{(3),(3,1)}$
$k = 1: \quad V^{(2),(3)} \oplus V^{(2),(2,1)} \oplus V^{(2),(3,1)} \oplus V^{(3),(3)} \oplus V^{(3),(2,1)}$
$k = 2: \quad 2V^{(1),(2)} \oplus 2V^{(1),(1,1)} \oplus 2V^{(1),(3)} \oplus 2V^{(1),(2,1)} \oplus V^{(1),(3,1)} \oplus 2V^{(2),(2)} \oplus 2V^{(2),(1,1)} \oplus 2V^{(2),(3)} \oplus 2V^{(2),(2,1)} \oplus V^{(3),(2)} \oplus V^{(3),(1,1)}$
$k = 3: \quad V^{\varnothing,(1)} \oplus V^{\varnothing,(2)} \oplus V^{\varnothing,(1,1)} \oplus V^{\varnothing,(3)} \oplus V^{\varnothing,(2,1)} \oplus 6V^{(1),(1)} \oplus 6V^{(1),(2)} \oplus 5V^{(1),(1,1)} \oplus 2V^{(1),(3)} \oplus 2V^{(1),(2,1)} \oplus 4V^{(2),(1)} \oplus 3V^{(2),(2)} \oplus 2V^{(2),(1,1)} \oplus V^{(3),(1)}$
$k = 4: \quad 3V^{\varnothing,\varnothing} \oplus 7V^{\varnothing,(1)} \oplus 6V^{\varnothing,(2)} \oplus 4V^{\varnothing,(1,1)} \oplus V^{\varnothing,(3)} \oplus V^{\varnothing,(2,1)} \oplus 6V^{(1),\varnothing} \oplus 12V^{(1),(1)} \oplus 5V^{(1),(2)} \oplus 4V^{(1),(1,1)} \oplus 3V^{(2),\varnothing} \oplus 3V^{(2),(1)}$
$k = 5: \quad 9V^{\varnothing,\varnothing} \oplus 14V^{\varnothing,(1)} \oplus 3V^{\varnothing,(2)} \oplus 2V^{\varnothing,(1,1)} \oplus 10V^{(1),\varnothing} \oplus 7V^{(1),(1)} \oplus V^{(2),\varnothing}$

Table 3: Table for $\lambda = (3,), \mu = (2,2), k = 6$

$k = 0: \quad V^{(3),(2,2)}$
$k = 1: \quad V^{(2),(2,1)} \oplus V^{(2),(2,2)} \oplus V^{(3),(2,1)}$
$k = 2: \quad 2V^{(1),(2)} \oplus 2V^{(1),(1,1)} \oplus 2V^{(1),(2,1)} \oplus V^{(1),(2,2)} \oplus 2V^{(2),(2)} \oplus 2V^{(2),(1,1)} \oplus 2V^{(2),(2,1)} \oplus V^{(3),(2)} \oplus V^{(3),(1,1)}$
$k = 3: \quad V^{\varnothing,(1)} \oplus V^{\varnothing,(2)} \oplus V^{\varnothing,(1,1)} \oplus V^{\varnothing,(2,1)} \oplus 6V^{(1),(1)} \oplus 5V^{(1),(2)} \oplus 5V^{(1),(1,1)} \oplus 2V^{(1),(2,1)} \oplus 4V^{(2),(1)} \oplus 2V^{(2),(2)} \oplus 2V^{(2),(1,1)} \oplus V^{(3),(1)}$
$k = 4: \quad 7V^{\varnothing,(1)} \oplus 4V^{\varnothing,(2)} \oplus 4V^{\varnothing,(1,1)} \oplus V^{\varnothing,(2,1)} \oplus 4V^{(1),\varnothing} \oplus 12V^{(1),(1)} \oplus 4V^{(1),(2)} \oplus 4V^{(1),(1,1)} \oplus 2V^{(2),\varnothing} \oplus 3V^{(2),(1)}$
$k = 5: \quad 8V^{\varnothing,\varnothing} \oplus 13V^{\varnothing,(1)} \oplus 2V^{\varnothing,(2)} \oplus 2V^{\varnothing,(1,1)} \oplus 8V^{(1),\varnothing} \oplus 5V^{(1),(1)} \oplus V^{(2),\varnothing}$

Table 4: Table for $\lambda = (3,)$, $\mu = (2,1,1)$, $k = 6$

$k=0: \quad V^{(3),(2,1,1)}$
$k=1: \quad V^{(2),(2,1)} \oplus V^{(2),(1,1,1)} \oplus V^{(2),(2,1,1)} \oplus V^{(3),(2,1)} \oplus V^{(3),(1,1,1)}$
$k=2: \quad 2V^{(1),(2)} \oplus 2V^{(1),(1,1)} \oplus 2V^{(1),(2,1)} \oplus 2V^{(1),(1,1,1)} \oplus V^{(1),(2,1,1)} \oplus 2V^{(2),(2)} \oplus 2V^{(2),(1,1)} \oplus 2V^{(2),(2,1)} \oplus 2V^{(2),(1,1,1)} \oplus V^{(3),(2)} \oplus V^{(3),(1,1)}$
$k=3: \quad V^{\varnothing,(1)} \oplus V^{\varnothing,(2)} \oplus V^{\varnothing,(1,1)} \oplus V^{\varnothing,(2,1)} \oplus V^{\varnothing,(1,1,1)} \oplus 6V^{(1),(1)} \oplus 5V^{(1),(2)} \oplus 6V^{(1),(1,1)} \oplus 2V^{(1),(2,1)} \oplus 2V^{(1),(1,1,1)} \oplus 4V^{(2),(1)} \oplus 2V^{(2),(2)} \oplus 3V^{(2),(1,1)} \oplus V^{(3),(1)}$
$k=4: \quad 7V^{\varnothing,(1)} \oplus 4V^{\varnothing,(2)} \oplus 6V^{\varnothing,(1,1)} \oplus V^{\varnothing,(2,1)} \oplus V^{\varnothing,(1,1,1)} \oplus 6V^{(1),\varnothing} \oplus 12V^{(1),(1)} \oplus 4V^{(1),(2)} \oplus 5V^{(1),(1,1)} \oplus 3V^{(2),\varnothing} \oplus 3V^{(2),(1)}$
$k=5: \quad 9V^{\varnothing,\varnothing} \oplus 14V^{\varnothing,(1)} \oplus 2V^{\varnothing,(2)} \oplus 3V^{\varnothing,(1,1)} \oplus 10V^{(1),\varnothing} \oplus 7V^{(1),(1)} \oplus V^{(2),\varnothing}$

Table 5: Table for $\lambda = (3,)$, $\mu = (1,1,1,1)$, $k = 6$

$k=0: \quad V^{(3),(1,1,1,1)}$
$k=1: \quad V^{(2),(1,1,1)} \oplus V^{(2),(1,1,1,1)} \oplus V^{(3),(1,1,1)}$
$k=2: \quad 2V^{(1),(1,1)} \oplus 2V^{(1),(1,1,1)} \oplus V^{(1),(1,1,1,1)} \oplus 2V^{(2),(1,1)} \oplus 2V^{(2),(1,1,1)} \oplus V^{(3),(1,1)}$
$k=3: \quad V^{\varnothing,(1)} \oplus V^{\varnothing,(1,1)} \oplus V^{\varnothing,(1,1,1)} \oplus 6V^{(1),(1)} \oplus 5V^{(1),(1,1)} \oplus 2V^{(1),(1,1,1)} \oplus 4V^{(2),(1)} \oplus 2V^{(2),(1,1)} \oplus V^{(3),(1)}$
$k=4: \quad 5V^{\varnothing,(1)} \oplus 4V^{\varnothing,(1,1)} \oplus V^{\varnothing,(1,1,1)} \oplus V^{(1),\varnothing} \oplus 10V^{(1),(1)} \oplus 4V^{(1),(1,1)} \oplus V^{(2),\varnothing} \oplus 2V^{(2),(1)} \oplus V^{(3),\varnothing}$
$k=5: \quad 6V^{\varnothing,\varnothing} \oplus 9V^{\varnothing,(1)} \oplus 2V^{\varnothing,(1,1)} \oplus 6V^{(1),\varnothing} \oplus 4V^{(1),(1)} \oplus 2V^{(2),\varnothing}$

Table 6: Table for $\lambda = (2,1)$, $\mu = (4,)$, $k = 6$

$k=0: \quad V^{(2,1),(4)}$
$k=1: \quad V^{(2),(3)} \oplus V^{(2),(4)} \oplus V^{(1,1),(3)} \oplus V^{(1,1),(4)} \oplus V^{(2,1),(3)}$
$k=2: \quad 2V^{(1),(2)} \oplus 2V^{(1),(3)} \oplus V^{(1),(4)} \oplus 2V^{(2),(2)} \oplus 2V^{(2),(3)} \oplus 2V^{(1,1),(2)} \oplus 2V^{(1,1),(3)} \oplus V^{(2,1),(2)}$
$k=3: \quad 4V^{\varnothing,(1)} \oplus 4V^{\varnothing,(2)} \oplus 2V^{\varnothing,(3)} \oplus 6V^{(1),(1)} \oplus 6V^{(1),(2)} \oplus 3V^{(1),(3)} \oplus 4V^{(2),(1)} \oplus 2V^{(2),(2)} \oplus 4V^{(1,1),(1)} \oplus 2V^{(1,1),(2)} \oplus V^{(2,1),(1)}$
$k=4: \quad 14V^{\varnothing,(1)} \oplus 6V^{\varnothing,(2)} \oplus V^{\varnothing,(3)} \oplus V^{(1),\varnothing} \oplus 12V^{(1),(1)} \oplus 5V^{(1),(2)} \oplus V^{(2),\varnothing} \oplus 2V^{(2),(1)} \oplus V^{(1,1),\varnothing} \oplus 2V^{(1,1),(1)}$
$k=5: \quad 8V^{\varnothing,\varnothing} \oplus 14V^{\varnothing,(1)} \oplus 3V^{\varnothing,(2)} \oplus 10V^{(1),\varnothing} \oplus 5V^{(1),(1)} \oplus V^{(2),\varnothing} \oplus V^{(1,1),\varnothing}$

Table 7: Table for $\lambda = (2,1)$, $\mu = (3,1)$, $k = 6$

$k=0: \quad V^{(2,1),(3,1)}$
$k=1:$ $V^{(2),(3)} \oplus V^{(2),(2,1)} \oplus V^{(2),(3,1)} \oplus V^{(1,1),(3)} \oplus V^{(1,1),(2,1)} \oplus V^{(1,1),(3,1)} \oplus V^{(2,1),(3)} \oplus V^{(2,1),(2,1)}$
$k=2:$ $2V^{(1),(2)} \oplus 2V^{(1),(1,1)} \oplus 2V^{(1),(3)} \oplus 2V^{(1),(2,1)} \oplus V^{(1),(3,1)} \oplus 2V^{(2),(2)} \oplus 2V^{(2),(1,1)} \oplus 2V^{(2),(3)} \oplus 2V^{(2),(2,1)} \oplus 2V^{(1,1),(2)} \oplus 2V^{(1,1),(1,1)} \oplus 2V^{(1,1),(3)} \oplus 2V^{(1,1),(2,1)} \oplus V^{(2,1),(2)} \oplus V^{(2,1),(1,1)}$
$k=3: \quad 4V^{\varnothing,(1)} \oplus 4V^{\varnothing,(2)} \oplus 4V^{\varnothing,(1,1)} \oplus 2V^{\varnothing,(3)} \oplus 2V^{\varnothing,(2,1)} \oplus 6V^{(1),(1)} \oplus 8V^{(1),(2)} \oplus 6V^{(1),(1,1)} \oplus 3V^{(1),(3)} \oplus 3V^{(1),(2,1)} \oplus 4V^{(2),(1)} \oplus 3V^{(2),(2)} \oplus 2V^{(2),(1,1)} \oplus 4V^{(1,1),(1)} \oplus 3V^{(1,1),(2)} \oplus 2V^{(1,1),(1,1)} \oplus V^{(2,1),(1)}$
$k=4: \quad 6V^{\varnothing,\varnothing} \oplus 16V^{\varnothing,(1)} \oplus 8V^{\varnothing,(2)} \oplus 6V^{\varnothing,(1,1)} \oplus V^{\varnothing,(3)} \oplus V^{\varnothing,(2,1)} \oplus 6V^{(1),\varnothing} \oplus 16V^{(1),(1)} \oplus 7V^{(1),(2)} \oplus 5V^{(1),(1,1)} \oplus 3V^{(2),\varnothing} \oplus 3V^{(2),(1)} \oplus 3V^{(1,1),\varnothing} \oplus 3V^{(1,1),(1)}$
$k=5: \quad 18V^{\varnothing,\varnothing} \oplus 20V^{\varnothing,(1)} \oplus 4V^{\varnothing,(2)} \oplus 3V^{\varnothing,(1,1)} \oplus 14V^{(1),\varnothing} \oplus 9V^{(1),(1)} \oplus V^{(2),\varnothing} \oplus V^{(1,1),\varnothing}$

Table 8: Table for $\lambda = (2,1)$, $\mu = (2,2)$, $k = 6$

$k=0: \quad V^{(2,1),(2,2)}$
$k=1: \quad V^{(2),(2,1)} \oplus V^{(2),(2,2)} \oplus V^{(1,1),(2,1)} \oplus V^{(1,1),(2,2)} \oplus V^{(2,1),(2,1)}$
$k=2: \quad 2V^{(1),(2)} \oplus 2V^{(1),(1,1)} \oplus 2V^{(1),(2,1)} \oplus V^{(1),(2,2)} \oplus 2V^{(2),(2)} \oplus 2V^{(2),(1,1)} \oplus 2V^{(2),(2,1)} \oplus 2V^{(1,1),(2)} \oplus 2V^{(1,1),(1,1)} \oplus 2V^{(1,1),(2,1)} \oplus V^{(2,1),(2)} \oplus V^{(2,1),(1,1)}$
$k=3: \quad 4V^{\varnothing,(1)} \oplus 4V^{\varnothing,(2)} \oplus 4V^{\varnothing,(1,1)} \oplus 2V^{\varnothing,(2,1)} \oplus 6V^{(1),(1)} \oplus 6V^{(1),(2)} \oplus 6V^{(1),(1,1)} \oplus 3V^{(1),(2,1)} \oplus 4V^{(2),(1)} \oplus 2V^{(2),(2)} \oplus 2V^{(2),(1,1)} \oplus 4V^{(1,1),(1)} \oplus 2V^{(1,1),(2)} \oplus 2V^{(1,1),(1,1)} \oplus V^{(2,1),(1)}$
$k=4: \quad 4V^{\varnothing,\varnothing} \oplus 16V^{\varnothing,(1)} \oplus 6V^{\varnothing,(2)} \oplus 6V^{\varnothing,(1,1)} \oplus V^{\varnothing,(2,1)} \oplus 4V^{(1),\varnothing} \oplus 16V^{(1),(1)} \oplus 5V^{(1),(2)} \oplus 5V^{(1),(1,1)} \oplus 2V^{(2),\varnothing} \oplus 3V^{(2),(1)} \oplus 2V^{(1,1),\varnothing} \oplus 3V^{(1,1),(1)}$
$k=5: \quad 14V^{\varnothing,\varnothing} \oplus 18V^{\varnothing,(1)} \oplus 3V^{\varnothing,(2)} \oplus 3V^{\varnothing,(1,1)} \oplus 12V^{(1),\varnothing} \oplus 7V^{(1),(1)} \oplus V^{(2),\varnothing} \oplus V^{(1,1),\varnothing}$

Table 9: Table for $\lambda = (2,1)$, $\mu = (2,1,1)$, $k = 6$

$k=0: \quad V^{(2,1),(2,1,1)}$
$k=1: \quad V^{(2),(2,1)} \oplus V^{(2),(1,1,1)} \oplus V^{(2),(2,1,1)} \oplus V^{(1,1),(2,1)} \oplus V^{(1,1),(1,1,1)} \oplus V^{(1,1),(2,1,1)} \oplus V^{(2,1),(2,1)} \oplus V^{(2,1),(1,1,1)}$
$k=2: \quad 2V^{(1),(2)} \oplus 2V^{(1),(1,1)} \oplus 2V^{(1),(2,1)} \oplus 2V^{(1),(1,1,1)} \oplus V^{(1),(2,1,1)} \oplus 2V^{(2),(2)} \oplus 2V^{(2),(1,1)} \oplus 2V^{(2),(2,1)} \oplus 2V^{(2),(1,1,1)} \oplus 2V^{(1,1),(2)} \oplus 2V^{(1,1),(1,1)} \oplus 2V^{(1,1),(2,1)} \oplus 2V^{(1,1),(1,1,1)} \oplus V^{(2,1),(2)} \oplus V^{(2,1),(1,1)}$
$k=3: \quad 4V^{\varnothing,(1)} \oplus 4V^{\varnothing,(2)} \oplus 4V^{\varnothing,(1,1)} \oplus 2V^{\varnothing,(2,1)} \oplus 2V^{\varnothing,(1,1,1)} \oplus 6V^{(1),(1)} \oplus 6V^{(1),(2)} \oplus 8V^{(1),(1,1)} \oplus 3V^{(1),(2,1)} \oplus 3V^{(1),(1,1,1)} \oplus 4V^{(2),(1)} \oplus 2V^{(2),(2)} \oplus 3V^{(2),(1,1)} \oplus 4V^{(1,1),(1)} \oplus 2V^{(1,1),(2)} \oplus 3V^{(1,1),(1,1)} \oplus V^{(2,1),(1)}$
$k=4: \quad 6V^{\varnothing,\varnothing} \oplus 16V^{\varnothing,(1)} \oplus 6V^{\varnothing,(2)} \oplus 8V^{\varnothing,(1,1)} \oplus V^{\varnothing,(2,1)} \oplus V^{\varnothing,(1,1,1)} \oplus 6V^{(1),\varnothing} \oplus 16V^{(1),(1)} \oplus 5V^{(1),(2)} \oplus 7V^{(1),(1,1)} \oplus 3V^{(2),\varnothing} \oplus 3V^{(2),(1)} \oplus 3V^{(1,1),\varnothing} \oplus 3V^{(1,1),(1)}$
$k=5: \quad 18V^{\varnothing,\varnothing} \oplus 20V^{\varnothing,(1)} \oplus 3V^{\varnothing,(2)} \oplus 4V^{\varnothing,(1,1)} \oplus 14V^{(1),\varnothing} \oplus 9V^{(1),(1)} \oplus V^{(2),\varnothing} \oplus V^{(1,1),\varnothing}$

Table 10: Table for $\lambda = (2,1)$, $\mu = (1,1,1,1)$, $k = 6$

$k=0: \quad V^{(2,1),(1,1,1,1)}$
$k=1: \quad V^{(2),(1,1,1)} \oplus V^{(2),(1,1,1,1)} \oplus V^{(1,1),(1,1,1)} \oplus V^{(1,1),(1,1,1,1)} \oplus V^{(2,1),(1,1,1)}$
$k=2: \quad 2V^{(1),(1,1)} \oplus 2V^{(1),(1,1,1)} \oplus V^{(1),(1,1,1,1)} \oplus 2V^{(2),(1,1)} \oplus 2V^{(2),(1,1,1)} \oplus 2V^{(1,1),(1,1)} \oplus 2V^{(1,1),(1,1,1)} \oplus V^{(2,1),(1,1)}$
$k=3: \quad 4V^{\varnothing,(1)} \oplus 4V^{\varnothing,(1,1)} \oplus 2V^{\varnothing,(1,1,1)} \oplus 6V^{(1),(1)} \oplus 6V^{(1),(1,1)} \oplus 3V^{(1),(1,1,1)} \oplus 4V^{(2),(1)} \oplus 2V^{(2),(1,1)} \oplus 4V^{(1,1),(1)} \oplus 2V^{(1,1),(1,1)} \oplus V^{(2,1),(1)}$
$k=4: \quad 14V^{\varnothing,(1)} \oplus 6V^{\varnothing,(1,1)} \oplus V^{\varnothing,(1,1,1)} \oplus V^{(1),\varnothing} \oplus 12V^{(1),(1)} \oplus 5V^{(1),(1,1)} \oplus V^{(2),\varnothing} \oplus 2V^{(2),(1)} \oplus V^{(1,1),\varnothing} \oplus 2V^{(1,1),(1)} \oplus V^{(2,1),\varnothing}$
$k=5: \quad 8V^{\varnothing,\varnothing} \oplus 14V^{\varnothing,(1)} \oplus 3V^{\varnothing,(1,1)} \oplus 10V^{(1),\varnothing} \oplus 5V^{(1),(1)} \oplus 2V^{(2),\varnothing} \oplus 2V^{(1,1),\varnothing}$

Table 11: Table for $\lambda = (1,1,1)$, $\mu = (4,)$, $k = 6$

$k=0: \quad V^{(1,1,1),(4)}$
$k=1: \quad V^{(1,1),(3)} \oplus V^{(1,1),(4)} \oplus V^{(1,1,1),(3)}$
$k=2: \quad 2V^{(1),(2)} \oplus 2V^{(1),(3)} \oplus V^{(1),(4)} \oplus 2V^{(1,1),(2)} \oplus 2V^{(1,1),(3)} \oplus V^{(1,1,1),(2)}$
$k=3: \quad V^{\varnothing,(1)} \oplus V^{\varnothing,(2)} \oplus V^{\varnothing,(3)} \oplus V^{\varnothing,(4)} \oplus 6V^{(1),(1)} \oplus 5V^{(1),(2)} \oplus 2V^{(1),(3)} \oplus 4V^{(1,1),(1)} \oplus 2V^{(1,1),(2)} \oplus V^{(1,1,1),(1)}$
$k=4: \quad 5V^{\varnothing,(1)} \oplus 4V^{\varnothing,(2)} \oplus 2V^{\varnothing,(3)} \oplus V^{(1),\varnothing} \oplus 10V^{(1),(1)} \oplus 4V^{(1),(2)} \oplus V^{(1,1),\varnothing} \oplus 2V^{(1,1),(1)}$
$k=5: \quad 6V^{\varnothing,\varnothing} \oplus 9V^{\varnothing,(1)} \oplus 3V^{\varnothing,(2)} \oplus 6V^{(1),\varnothing} \oplus 4V^{(1),(1)} \oplus V^{(1,1),\varnothing}$

Table 12: Table for $\lambda = (1,1,1)$, $\mu = (3,1)$, $k = 6$

$k=0: \quad V^{(1,1,1),(3,1)}$
$k=1: \quad V^{(1,1),(3)} \oplus V^{(1,1),(2,1)} \oplus V^{(1,1),(3,1)} \oplus V^{(1,1,1),(3)} \oplus V^{(1,1,1),(2,1)}$
$k=2: \quad 2V^{(1),(2)} \oplus 2V^{(1),(1,1)} \oplus 2V^{(1),(3)} \oplus 2V^{(1),(2,1)} \oplus V^{(1),(3,1)} \oplus 2V^{(1,1),(2)} \oplus 2V^{(1,1),(1,1)} \oplus 2V^{(1,1),(3)} \oplus 2V^{(1,1),(2,1)} \oplus V^{(1,1,1),(2)} \oplus V^{(1,1,1),(1,1)}$
$k=3: \quad V^{\varnothing,(1)} \oplus V^{\varnothing,(2)} \oplus V^{\varnothing,(1,1)} \oplus V^{\varnothing,(3)} \oplus V^{\varnothing,(2,1)} \oplus V^{\varnothing,(3,1)} \oplus 6V^{(1),(1)} \oplus 6V^{(1),(2)} \oplus 5V^{(1),(1,1)} \oplus 2V^{(1),(3)} \oplus 2V^{(1),(2,1)} \oplus 4V^{(1,1),(1)} \oplus 3V^{(1,1),(2)} \oplus 2V^{(1,1),(1,1)} \oplus V^{(1,1,1),(1)}$
$k=4: \quad 7V^{\varnothing,(1)} \oplus 6V^{\varnothing,(2)} \oplus 4V^{\varnothing,(1,1)} \oplus 2V^{\varnothing,(3)} \oplus 2V^{\varnothing,(2,1)} \oplus 6V^{(1),\varnothing} \oplus 12V^{(1),(1)} \oplus 5V^{(1),(2)} \oplus 4V^{(1),(1,1)} \oplus 3V^{(1,1),\varnothing} \oplus 3V^{(1,1),(1)}$
$k=5: \quad 9V^{\varnothing,\varnothing} \oplus 14V^{\varnothing,(1)} \oplus 4V^{\varnothing,(2)} \oplus 3V^{\varnothing,(1,1)} \oplus 10V^{(1),\varnothing} \oplus 7V^{(1),(1)} \oplus V^{(1,1),\varnothing}$

Table 13: Table for $\lambda = (1,1,1)$, $\mu = (2,2)$, $k = 6$

$k=0: \quad V^{(1,1,1),(2,2)}$
$k=1: \quad V^{(1,1),(2,1)} \oplus V^{(1,1),(2,2)} \oplus V^{(1,1,1),(2,1)}$
$k=2: \quad 2V^{(1),(2)} \oplus 2V^{(1),(1,1)} \oplus 2V^{(1),(2,1)} \oplus V^{(1),(2,2)} \oplus 2V^{(1,1),(2)} \oplus 2V^{(1,1),(1,1)} \oplus 2V^{(1,1),(2,1)} \oplus V^{(1,1,1),(2)} \oplus V^{(1,1,1),(1,1)}$
$k=3: \quad V^{\varnothing,(1)} \oplus V^{\varnothing,(2)} \oplus V^{\varnothing,(1,1)} \oplus V^{\varnothing,(2,1)} \oplus V^{\varnothing,(2,2)} \oplus 6V^{(1),(1)} \oplus 5V^{(1),(2)} \oplus 5V^{(1),(1,1)} \oplus 2V^{(1),(2,1)} \oplus 4V^{(1,1),(1)} \oplus 2V^{(1,1),(2)} \oplus 2V^{(1,1),(1,1)} \oplus V^{(1,1,1),(1)}$
$k=4: \quad 7V^{\varnothing,(1)} \oplus 4V^{\varnothing,(2)} \oplus 4V^{\varnothing,(1,1)} \oplus 2V^{\varnothing,(2,1)} \oplus 4V^{(1),\varnothing} \oplus 12V^{(1),(1)} \oplus 4V^{(1),(2)} \oplus 4V^{(1),(1,1)} \oplus 2V^{(1,1),\varnothing} \oplus 3V^{(1,1),(1)}$
$k=5: \quad 8V^{\varnothing,\varnothing} \oplus 13V^{\varnothing,(1)} \oplus 3V^{\varnothing,(2)} \oplus 3V^{\varnothing,(1,1)} \oplus 8V^{(1),\varnothing} \oplus 5V^{(1),(1)} \oplus V^{(1,1),\varnothing}$

Table 14: Table for $\lambda = (1,1,1)$, $\mu = (2,1,1)$, $k = 6$

$k=0: \quad V^{(1,1,1),(2,1,1)}$
$k=1: \quad V^{(1,1),(2,1)} \oplus V^{(1,1),(1,1,1)} \oplus V^{(1,1),(2,1,1)} \oplus V^{(1,1,1),(2,1)} \oplus V^{(1,1,1),(1,1,1)}$
$k=2: \quad 2V^{(1),(2)} \oplus 2V^{(1),(1,1)} \oplus 2V^{(1),(2,1)} \oplus 2V^{(1),(1,1,1)} \oplus V^{(1),(2,1,1)} \oplus 2V^{(1,1),(2)} \oplus 2V^{(1,1),(1,1)} \oplus 2V^{(1,1),(2,1)} \oplus 2V^{(1,1),(1,1,1)} \oplus V^{(1,1,1),(2)} \oplus V^{(1,1,1),(1,1)}$
$k=3: \quad V^{\varnothing,(1)} \oplus V^{\varnothing,(2)} \oplus V^{\varnothing,(1,1)} \oplus V^{\varnothing,(2,1)} \oplus V^{\varnothing,(1,1,1)} \oplus V^{\varnothing,(2,1,1)} \oplus 6V^{(1),(1)} \oplus 5V^{(1),(2)} \oplus 6V^{(1),(1,1)} \oplus 2V^{(1),(2,1)} \oplus 2V^{(1),(1,1,1)} \oplus 4V^{(1,1),(1)} \oplus 2V^{(1,1),(2)} \oplus 3V^{(1,1),(1,1)} \oplus V^{(1,1,1),(1)}$
$k=4: \quad 3V^{\varnothing,\varnothing} \oplus 7V^{\varnothing,(1)} \oplus 4V^{\varnothing,(2)} \oplus 6V^{\varnothing,(1,1)} \oplus 2V^{\varnothing,(2,1)} \oplus 2V^{\varnothing,(1,1,1)} \oplus 6V^{(1),\varnothing} \oplus 12V^{(1),(1)} \oplus 4V^{(1),(2)} \oplus 5V^{(1),(1,1)} \oplus 3V^{(1,1),\varnothing} \oplus 3V^{(1,1),(1)}$
$k=5: \quad 9V^{\varnothing,\varnothing} \oplus 14V^{\varnothing,(1)} \oplus 3V^{\varnothing,(2)} \oplus 4V^{\varnothing,(1,1)} \oplus 10V^{(1),\varnothing} \oplus 7V^{(1),(1)} \oplus V^{(1,1),\varnothing}$

Table 15: Table for $\lambda = (1,1,1)$, $\mu = (1,1,1,1)$, $k = 6$

$k=0: \quad V^{(1,1,1),(1,1,1,1)}$
$k=1: \quad V^{(1,1),(1,1,1)} \oplus V^{(1,1),(1,1,1,1)} \oplus V^{(1,1,1),(1,1,1)}$
$k=2: \quad 2V^{(1),(1,1)} \oplus 2V^{(1),(1,1,1)} \oplus V^{(1),(1,1,1,1)} \oplus 2V^{(1,1),(1,1)} \oplus 2V^{(1,1),(1,1,1)} \oplus V^{(1,1,1),(1,1)}$
$k=3: \quad V^{\varnothing,(1)} \oplus V^{\varnothing,(1,1)} \oplus V^{\varnothing,(1,1,1)} \oplus V^{\varnothing,(1,1,1,1)} \oplus 6V^{(1),(1)} \oplus 5V^{(1),(1,1)} \oplus 2V^{(1),(1,1,1)} \oplus 4V^{(1,1),(1)} \oplus 2V^{(1,1),(1,1)} \oplus V^{(1,1,1),(1)}$
$k=4: \quad V^{\varnothing,\varnothing} \oplus 5V^{\varnothing,(1)} \oplus 4V^{\varnothing,(1,1)} \oplus 2V^{\varnothing,(1,1,1)} \oplus V^{(1),\varnothing} \oplus 10V^{(1),(1)} \oplus 4V^{(1),(1,1)} \oplus V^{(1,1),\varnothing} \oplus 2V^{(1,1),(1)} \oplus V^{(1,1,1),\varnothing}$
$k=5: \quad 7V^{\varnothing,\varnothing} \oplus 9V^{\varnothing,(1)} \oplus 3V^{\varnothing,(1,1)} \oplus 6V^{(1),\varnothing} \oplus 4V^{(1),(1)} \oplus 2V^{(1,1),\varnothing}$